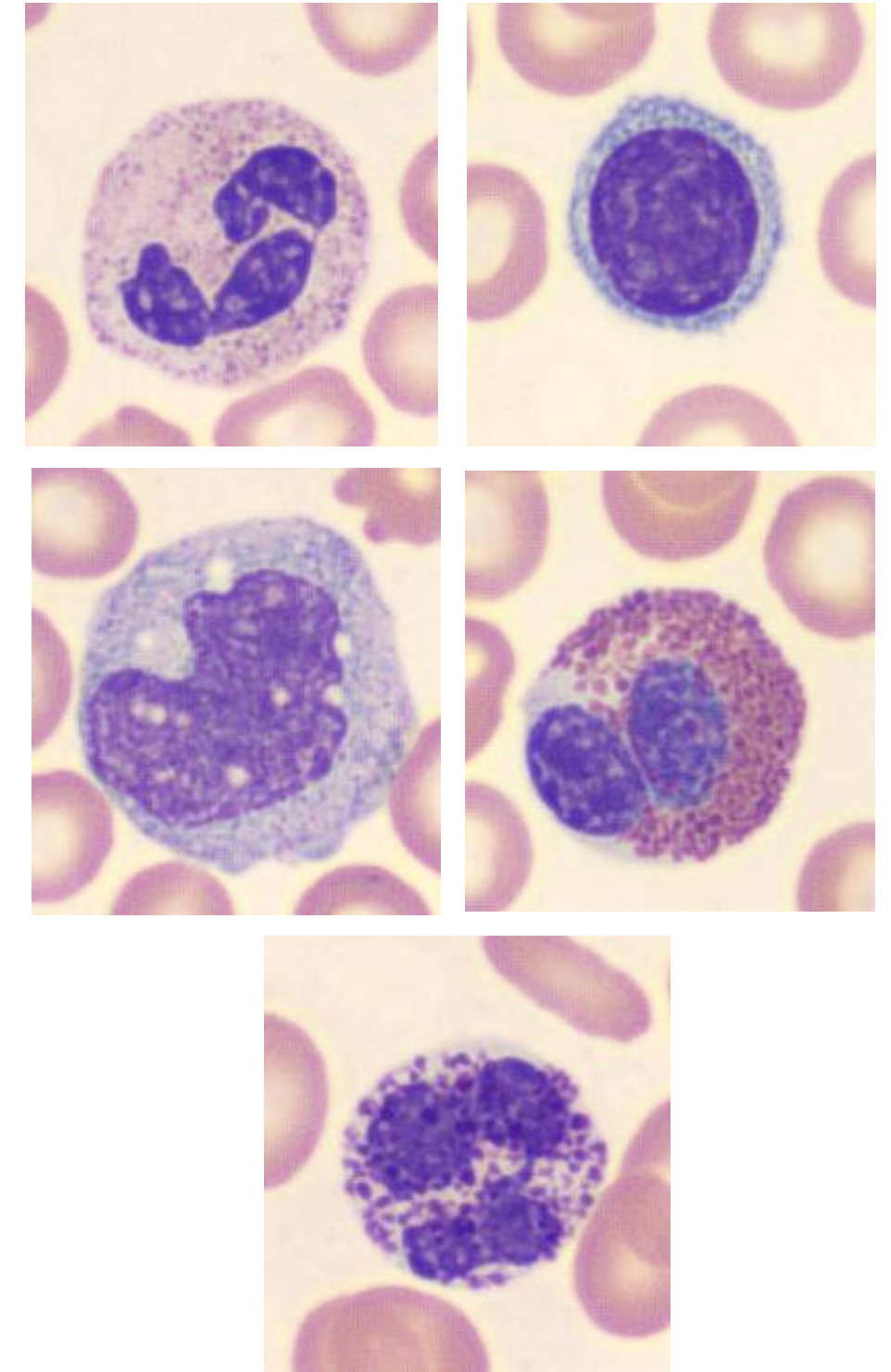


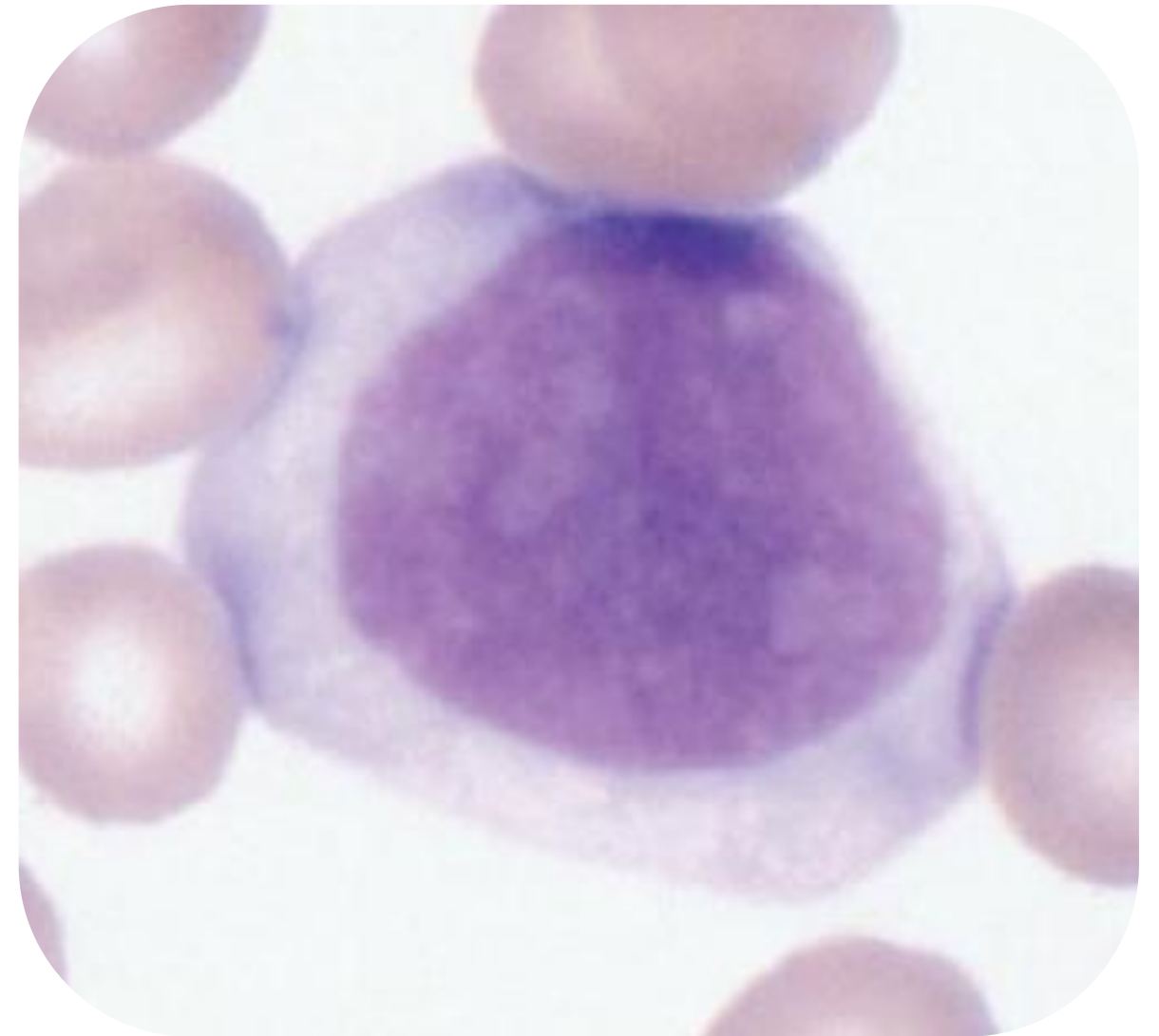
Monocytes

- ~2-10% of total WBC
- Host defense in tissues as a macrophage
- Morphologically recognizable stages
 - Monoblast
 - Promonocyte
 - Monocyte



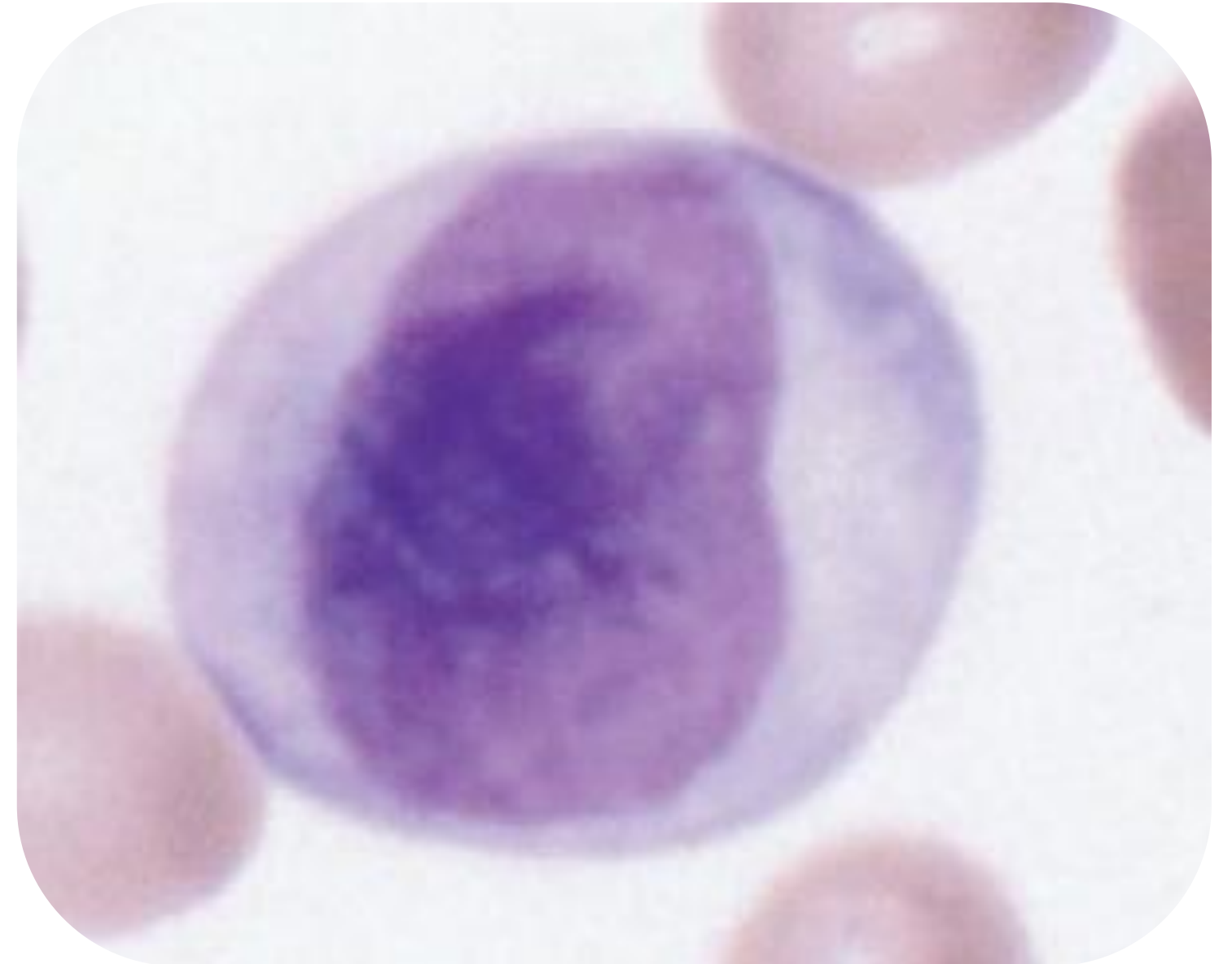
Blast (Monoblast)

- 12-20 μm
- Nucleus
 - Ovoid/round, can be folded/indented
 - Pale blue-purple, fine/lacy chromatin
- Cytoplasm
 - Abundant, agranular, blue-gray
- Cytochemical and surface
 - NSE+, CD 14+



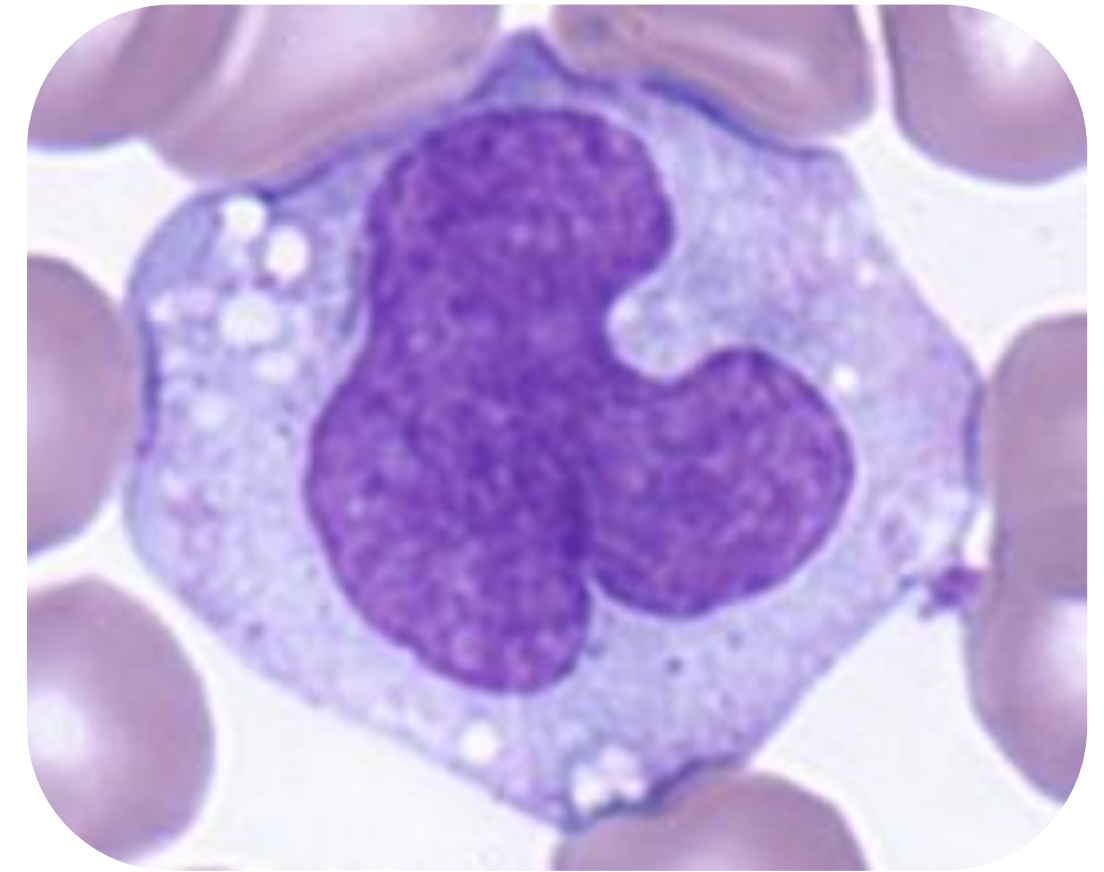
Blast (Promonocyte)

- 12-20 μm
- Nucleus
 - Irregular/indented
 - Fine/coarse chromatin
- Cytoplasm
 - Abundant, blue-gray
 - Azurophilic granules possible



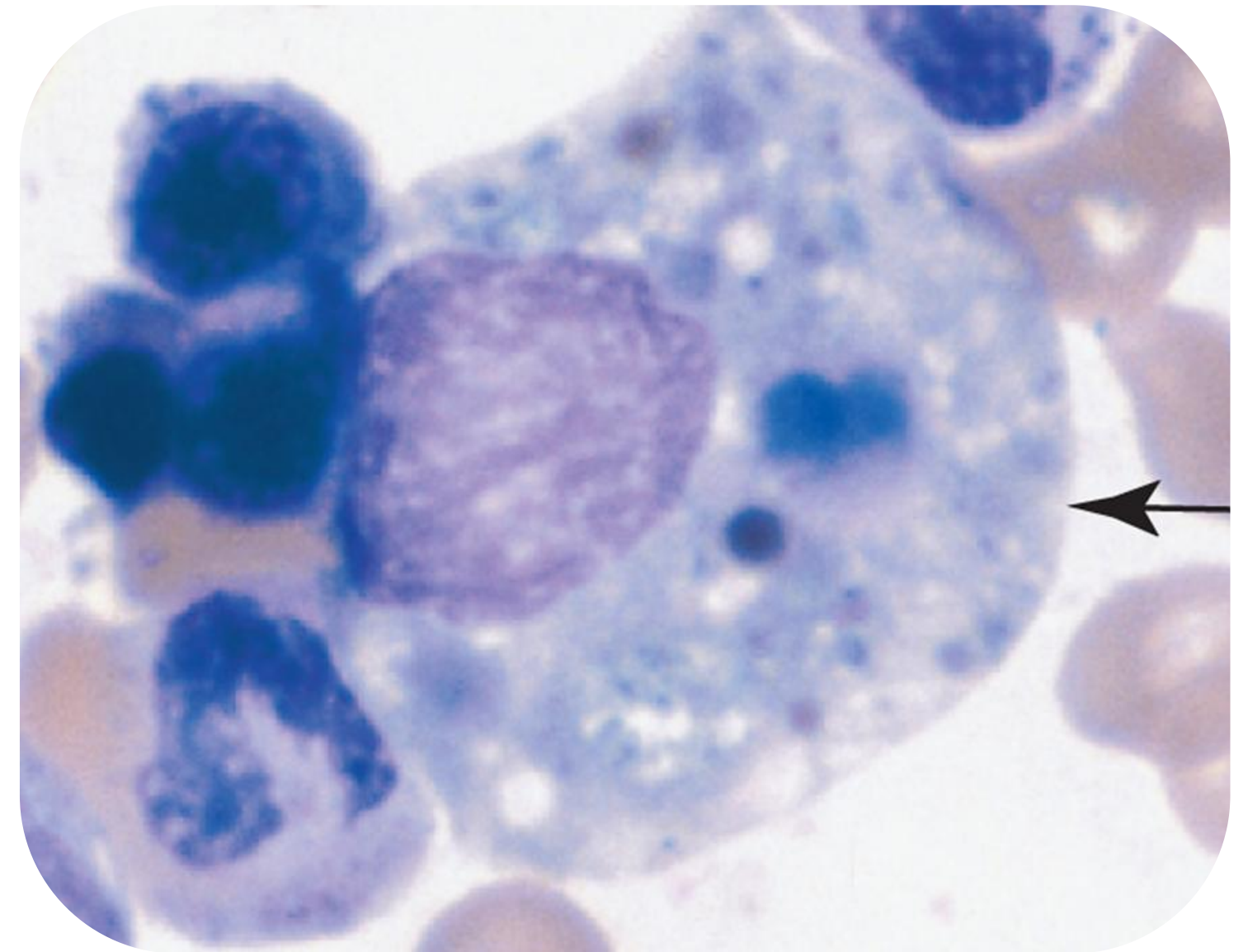
Monocyte

- 12-20 μm
- Nucleus
 - Horseshoe/bean shaped with many folds
 - Chromatin loose and linear, lacy
- Cytoplasm
 - Blue-gray
 - Dust-like granules = ground glass appearance
 - Vacuoles frequently seen



Monocyte → Macrophage

- Leaves PB to tissues
- 15-80 μm
- Specific names to location
 - Tissue = histiocyte
 - Liver = Kupffer cell
 - Lungs = alveolar macrophage
 - Skin = Langerhans cells
 - Brain = microglial cells

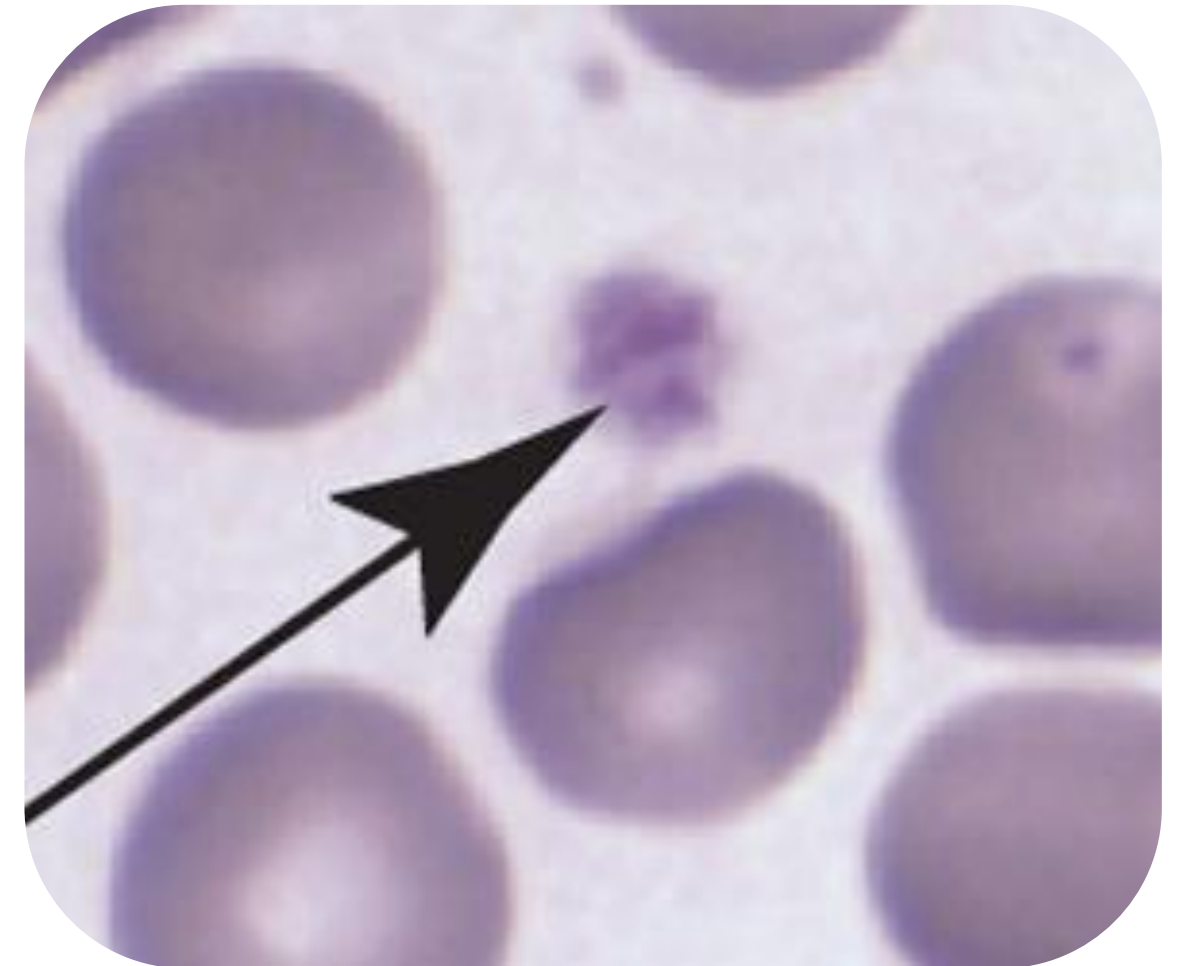


Monocyte Kinetics/Function

- Diapedesis = passage into tissue after ~8 hours
- Marginating pool ~3x circulating
- Active in innate and adaptive immune response
 - T cell activated → Ingest + kill microorganisms
 - Attach to tumor cells and kill
- Macrophages and phagocytosis

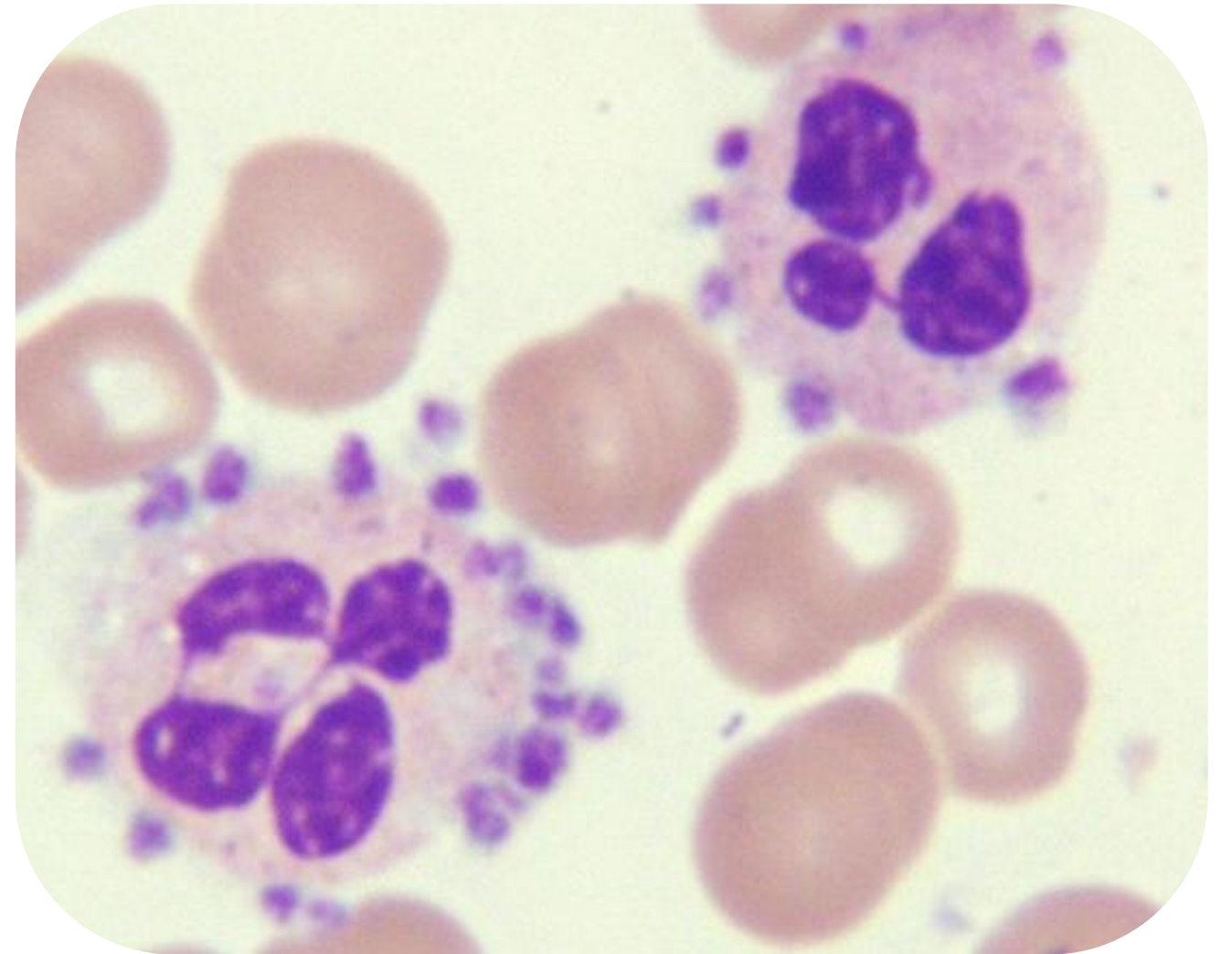
Platelets Kinetics

- Anucleated fragments of cytoplasm
- Lifespan in PB ~ 7-10 days
 - $[PB] = 150 - 450 \times 10^3/\mu L$
- Two pools = 2/3 circulating + 1/3 sequestered in spleen
- Reticulated platelets
 - Immature
 - ~5-10%
 - $\uparrow$ associated with $\uparrow$ platelet destruction



Platelet Morphology

- 2-3 μm , round/oval
- Giant platelets
 - $> 5 \mu\text{m}$
 - MPD, recovery
- EDTA can cause platelet clumping or satellitosis



Platelet Function

- Primary defense against bleeding
 - Adhesion
 - Subendothelial vWF + PLT GPIb
 - Binds exposed collagen at high shear rate
 - Shape change (contraction)
 - Secrete substances to recruit other platelets
 - Aggregation
 - Fibrinogen + PLT GPIIb/IIIa
 - Bridge connection to other platelets